

Mike Hall

Senior / Lead Software Engineer, Ruby on Rails

Chicago, IL • mike@just3ws.com • (847) 877-3825 • just3ws.com • linkedin.com/in/just3ws/ • github.com/just3ws

Senior software engineer focused on Ruby on Rails infrastructure, PostgreSQL, legacy modernization, and production reliability. Delivers business value through complex state machine refactoring, database optimization, asynchronous processing, third-party integrations, and disciplined testing.

EXPERIENCE & LEADERSHIP

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Target Focus: Architected and delivered an instant loan disbursement pipeline (funding to debit cards in minutes) and refactored multi-step Rails workflow state machines, eliminating lateral state corruptions across high-volume customer prequalification pipelines.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Target Focus: Sequenced zero-downtime Ruby and Rails version upgrades, optimized MySQL full-text search indices, and built batch Sidekiq/SendGrid delivery pipelines.

- **Search Infrastructure Modernization:** Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Rails Upgrades:** Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Target Focus: Introduced testable query patterns and caching into a tightly coupled PHP/MySQL CRM backend and extended the Ember.js frontend test suite across contact management workflows.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Senior Software Developer

Upcity | Remote | October 2013 - February 2014

Target Focus: Integrated Chargify recurring billing and standardized multi-VM local development environments via Vagrant.

- Integrated the Chargify subscription payment platform, launching self-service recurring billing and account management.
- Standardized local development environments using multi-VM Vagrant orchestration, eliminating developer environment drift against production.

Senior Software Developer

Viewpoints | Chicago, IL | May 2013 - October 2013

Target Focus: Implemented custom tracking analytics and automated staging/production deployment pipelines for high-traffic consumer web products.

- Implemented custom tracking and analytics tools, enabling targeted monetization and direct platform revenue.
- Standardized automated testing and staging environments, establishing reliable continuous integration across the engineering team.
- Streamlined AWS deployment pipelines and server images, eliminating deployment downtime during peak consumer traffic spikes.

SELECTED PRODUCTION PROJECTS

Principal Architect

WWWorkRemote | Independent Project | January 2021 - Present

Rails 8 platform for multi-source job ingestion, semantic matching, and application automation, built as a working laboratory for local-first LLM orchestration and treating scraped third-party text as hostile input.

- **Multi-Source Data Ingestion:** Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.
- **Semantic Search & Embeddings:** Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.
- **Local-First AI Orchestration:** Designed provider-agnostic LLM orchestration layer (llama.cpp/Ollama with hosted Claude fallback), enforcing local-first inference for sensitive data with per-call model routing.
- **Prompt Injection Defense:** Built a 4-stage guardrails pipeline screening untrusted scraped text via weighted heuristics and instruction-density scoring before model ingestion.
- **Browser Extension & Lifecycle Capture:** Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures full application lifecycles.
- **Architectural Constraint Injection:** Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and security contracts.

Principal Architect

Phalanx Duel | Independent Project | January 2022 - Present

Designing and building a real-time tactical game platform as a current hands-on laboratory for deterministic systems, product architecture, and controlled AI-assisted engineering.

- Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.
- Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.
- Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | January 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- **Bidirectional Knowledge Server:** Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- **Telemetry-Driven Diagnosis:** Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- **Context as a Budgeted Resource:** Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

CORE SKILLS & COMPETENCIES

01 • Ruby on Rails (Full-Lifecycle 2.x through 8.x)	02 • PostgreSQL & MySQL Query Optimization & Indexing	03 • Asynchronous Processing (Sidekiq, Resque, Redis)	04 • State Machine Refactoring & Data Integrity
05 • Third-Party API & Webhook Integrations	06 • Test-Driven Development (RSpec, Capybara, FactoryBot)	07 • Docker, Linux, Git & Zero-Downtime Deployment	08 • Fast Ramp-Up & Autonomous IC Execution

ADDITIONAL EXPERIENCE

- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Senior Software Developer, BenchPrep** | March 2017 - February 2018 | Chicago, IL: Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- **Open-Source Transition Lead, Coderwall** | January 2014 - December 2014 | Remote: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Principal Consultant, Tandem** | August 2018 - August 2018 | Chicago, IL: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.

EARLIER EXPERIENCE

Selected earlier experience

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that inform my leadership today.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.